Design Proposal for Interferometry Using Mach-Zehnder Interferometers and Ring Resonators

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1 Introduction

Silicon photonics is a growing field that aims to replicate the vesatility and applications of free-space or fibre-based optics, using mature silicon fabrication processes, to miniaturise systems: an optical set-up that may require an entire optical bench in free space could be achieved on a silicon chip centimetres or milimetres in size. This is achieved through micrometre and nanometre-scale waveguide-based passive components, such as beamsplitters, crossings and couplers, in concert with active components such as phase changers, and on-chip light sources and detectors. In this article, I propose three different designs for performing interferometry: a Mach-Zehnder interferometer (MZI), a coupled ring resonator-based optical reflector, and a ring resonator-based demultiplexer (demux). These three devices will allow calculation of the group index in a silicon waveguide.

2 Theory

In this section I will outline the theory of the MZI and the ring resonator, and show how group index can be calculated through measurments of these devices.

2.1 Mach-Zehnder interferometer

Simply put, a MZI is two coupled 2x2 splitters. The incident light is split at the first splitter, and then interferes at the second splitter, changing the amplitude at each output. By introducing a path length difference in the intermediate arms of the MZI, creating a so-called "unbalanced" MZI, the fraction of the transmitted light from each output changes as a function of the phase shift due to the path length difference.

The electric field of the light transmitted to the intermediate arms of the MZI by the initial splitter is given by: $E_1 = E_2 = \frac{E_i}{\sqrt{2}}$. The light propagates (with propagation constant $\beta = 2\pi n_{eff}/\lambda$) down waveguides with path length L_1 and L_2. These meet at the second splitter. At the input to the second splitter, the fields are given by:

$$\begin{array}{rcl} E_1 & \rightarrow & E_1 e^{-i\beta L_1} \\ E_2 & \rightarrow & E_2 e^{-i\beta L_2} \end{array}$$

The field after the splitter on one of the outputs is

$$E_o = \frac{E_1 + E_2}{2}$$

and the intensity is

$$I_o = \frac{(E_1^* E_1 + E_2^* E_2 + E_1^* E_2 + E_2^* E_1)}{4}$$

The $E_i^* E_j$ terms with $i = j$ give a contribution of $I_i/2$, whereas the interference terms where $i \neq j$ introduce a sinusoidal term dependent on the path length difference $\Delta L = L_1 - L_2$. This results in output intensity:

$$\begin{array}{rcl} I_o & = & \frac{I_i}{2}(1 + \cos(\beta \Delta L) \\ & \propto & (1 + \cos(2\pi n_{eff} \Delta L / \lambda) \end{array}$$

Thus by sweeping wavelength, the intensity will oscillate between maximum and zero. The peak-to-peak distance ΔL is known as the free spectral range (FSR). The group index can be calculated from the FSR. The phase shift in the transfer function $\delta = \beta \Delta L$, and the phase difference between peaks is

$$\begin{array}{rcl} \Delta\delta & = & \delta_m - \delta_{m+1} = 2\pi \\ & = & \beta_m \Delta L - \beta_{m+1} \Delta L \\ & = & \Delta\beta \Delta L \\ \Delta\beta & = & \frac{2\pi}{\Delta L} \end{array}$$

Assuming β varies linearly wiht λ:

$$\Delta\beta = -\frac{d\beta}{d\lambda} \Delta\lambda = \frac{2\pi}{\Delta L}$$

Calculating $d\beta/d\lambda$:

$$\frac{d\beta}{d\lambda} = \frac{2\pi}{\lambda}\frac{dn}{d\lambda} + 2\pi n(-\lambda^{-2}) = -\frac{2\pi}{\lambda^2}\left(n - \frac{dn}{d\lambda}\lambda\right)$$

Solving for $\Delta\lambda$, we get

$$\Delta\lambda = \frac{\lambda^2}{\Delta L(n - \lambda(dn/d\lambda)}$$

where

$$n - \lambda\frac{dn}{d\lambda} = n_g.$$

2.2 Ring resonator

A ring resonator is a circular waveguide that acts as an optical cavity. Light is fed into it from an adjacent waveguide via the evanescent field, and round-trip interference for light of different wavelengths means that only certain modes are supported in the resonator. The optical path length of a ring resonator is given by $2\pi r n_{eff}$ where r is the radius of the ring. For resonance of a given mode, the optical path length must be equal to an integer number of wavelengths: $m\lambda_m$. For a given size of ring resonator, the FSR is the difference in wavelength between modes m and $m \pm 1$, i.e. the modes where the ring supports m and $m \pm 1$ wavelengths. This corresponds to a phase difference of 2π over the full ring round trip, so the FSR can be calculated similarly to the MZI: $\Delta\beta L = 2\pi, \beta = 2\pi n(\lambda)/\lambda, L = 2\pi r$. By expanding $\Delta\beta$ as a Taylor series, the FSR is given as

$$\Delta\lambda = \frac{\lambda^2}{n_g L} = \frac{\lambda^2}{2\pi r n_g}$$

where the group index $n_g = n(\lambda) - \lambda\frac{\partial n}{\partial\lambda}$.

2.2.1 Optical reflector

This device is given by two identical ring resonators in series along a waveguide that are in turn coupled together. Modes that couple into the first ring in the anticlockwise direction can in turn couple into the second ring in the clockwise direction, and then couple into the base waveguide propagating backwards. In turn, modes that couple into the second ring resonator from the waveguide can couple into the second ring and then couple back into the waveguide, also propagating backwards. Thus the device acts as a wavelength-selective filter/mirror, reflecting a portion of the incident light back along the waveguide, resulting in a reduction in the transmitted power at the ring modes. The FSR can be extracted from the transmitted spectrum as a series of dips.

2.2.2 Demultiplexer

This device is made up of a series of cascaded ring resonators that are coupled to two waveguides at opposite sides of the ring. Resonant modes of the first ring are coupled to the second waveguide output, allowing collection of light of those resonant modes. By changing the radii of the subsequent ring resonators, successive sets of modes can be picked off the incident light and coupled out to detectors. This allows a number of ring resonators to be measured from one input, each with its own FSR, and thus each allowing another calculation of the group index.

3 Modelling and simulation

Modelling of the waveguides and MZIs was performed using Lumerical MODE and INTERCONNECT. Modelling of ring resonators performed using the 2.5D varFDTD mode in Lumerical MODE - full 3D FDTD simulations were not performed due to the size of the structures.

3.1 TE waveguide mode

The waveguide geometry was standard for 1.55μm wavelength: 0.5 μm width and 0.22 μm thickness. Mode calculations were performed using MODE's fiite difference eigenmode solver and showed fundamental mode indices of 2.448 for TE amd 1.76 for TM at 1.55 μm. A frequency sweep from 1.5 to 1.6 microns show a strongly linear response for the real part between 2.50 and 2.39. The group index over the same range rises from 4.17 to 4.28.

MODE simulations were also perfomed for bent waveguides with radius around 20 microns - similar to those of the ring resonators (assuming m of around 200). The values for effective and group index from these simulation show very good agreement with those of the straight waveguides.

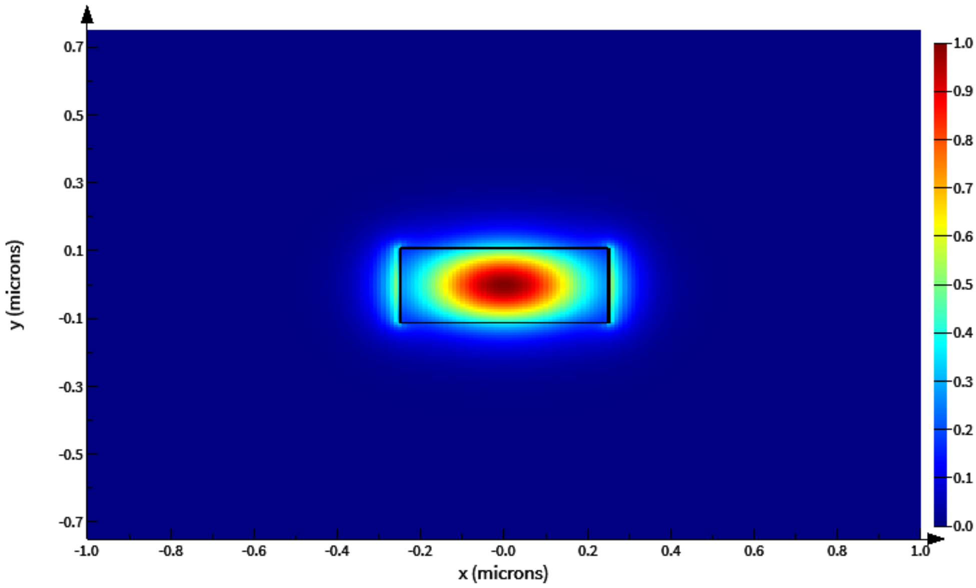
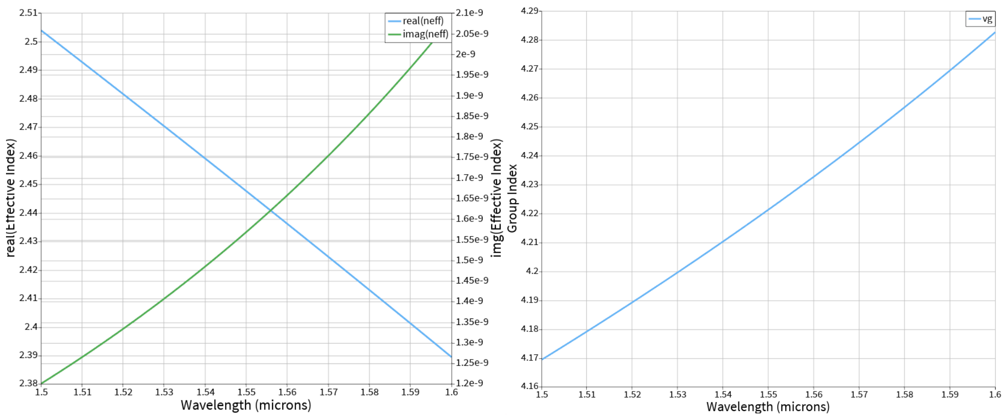


Figure 1: Fundamental TE mode in waveguide

3.2 MZI

Three designs for MZIs are included in the layout, with different ΔL values, outlined in Table. 1. The MZI circuits from the .gds file were simulated in Lumerical INTERCONNECT via KLayout, and mean group index values were calculated from the FSRs.



(a) (b)

Figure 2: (a) plot of effective index (b) plot of group index. Both from MODE simulation.

MZI	ΔL (μm)	simulated n_g
1	244.4	4.22 $\pm$ 0.08
2	103.5	4.26 $\pm$ 0.06
3	396.3	4.21 $\pm$ 0.03

Table 1: Table of MZI path lengths and simulated group indices (n_g).

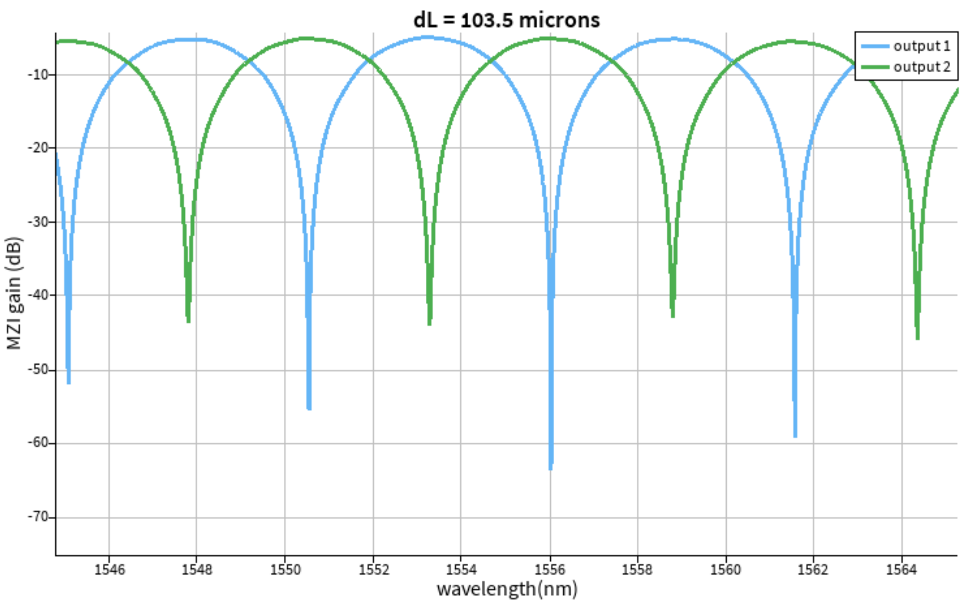


Figure 3: Plot of gain to both outputs of MZI circuit. Distance between peaks is FSR.

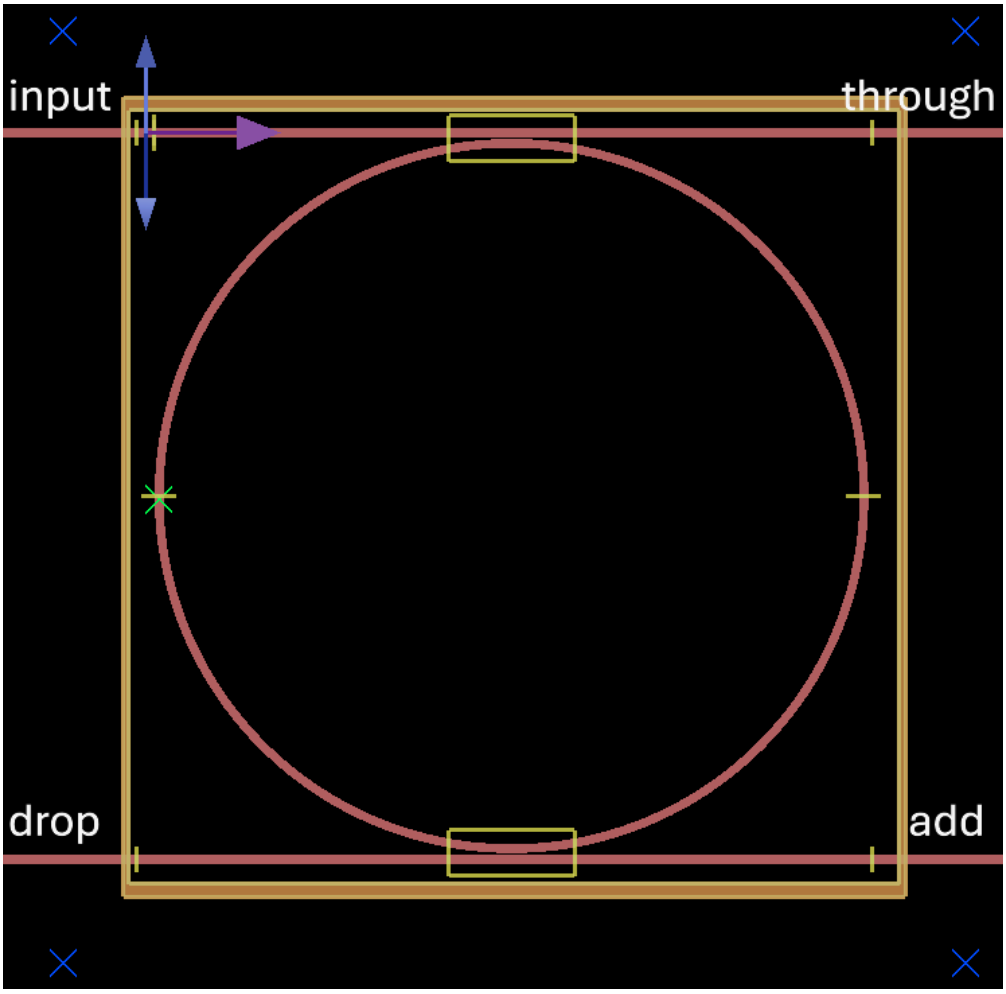
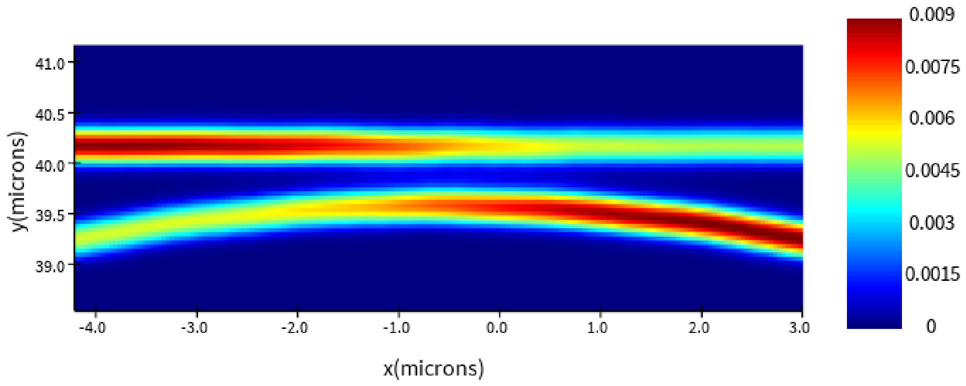


Figure 4: Lumerical MODE view of dual-bus ring. Input/output ports are labelled.

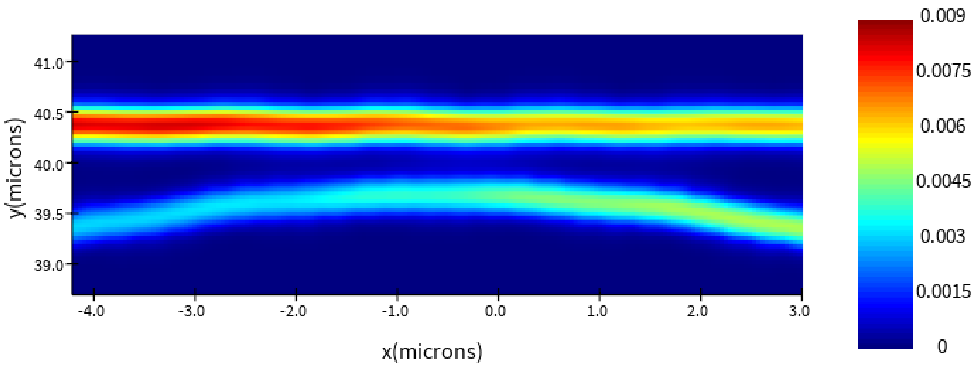
3.3 Dual-bus ring resonator

This component is used in the optical demultiplexer - see Fig. 4. The dual-bus ring is coupled to two waveguides at opposite sides of the ring.[1] Light at the correct wavelength, given by the resonance relationship above, can couple into the ring and circulate, and in turn couple to the second waveguide and exit through 'drop'. The coupling is determined by the evanescent field established by the waveguide - in simple terms, the closer the waveguide is to the ring, and the longer that the waveguide is close to the ring, the higher the degree of coupling.

[1]The names of the ports are chosen due to their relationship to the input waveguide: 'through' carries light that does not couple to the ring. Light that couples to the ring from 'input' can then leave the ring and exit through 'drop' - light is dropped from the through waveguide. If light were incident at 'add' and coupled to the ring, it would be 'added' to the input-through waveguide.



(a)

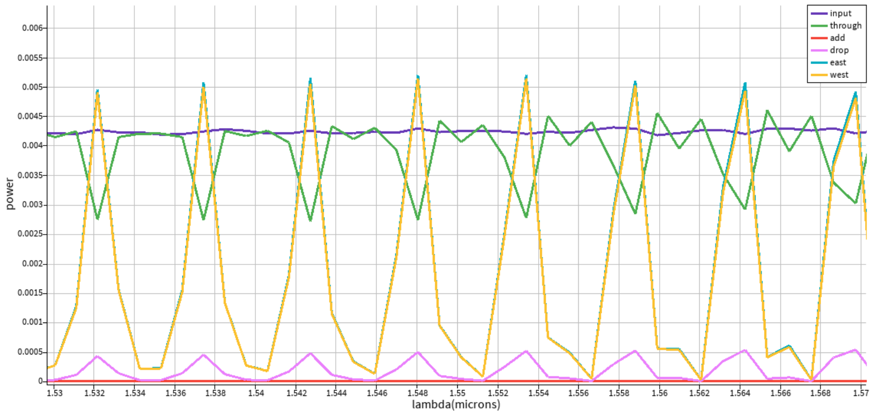


(b)

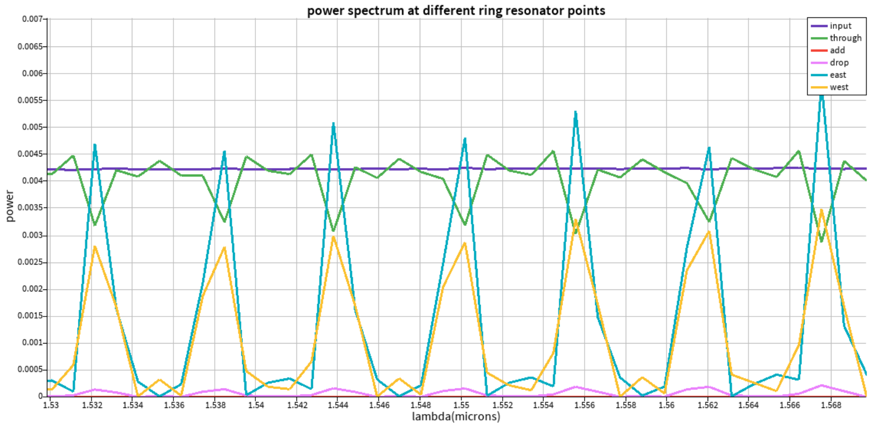
Figure 5: FDTD simulation detail of ring resonator coupler for gasp of (a) 100nm and (b) 200nm. Changing the gap gives a small ($\tilde{2}$nm) shift to the resonant wavelength: this may be due to interference at the couplers affecting the optical path length in the ring.

In my designs, I have included two ring-waveguide gaps: 100nm and 200nm. Performing varFDTD simulations, the difference in coupling strength can be seen for the different gaps: see Fig. 5. The smaller gap clearly allows more light to couple into the ring. This is further demonstrated in plots of the power at different points on the ring, as seen in Fig. 6. A smaller gap results in, at resonance, lower power to the through port, equal power at opposite points in the ring, and greater power to the drop port.

Three ring radii are included in the multiplexer, corresponding to three sets of resonant modes. The radii are calculated to support three chosen wavelengths with $m = 200$. These are given in Table. 2.



(a)



(b)

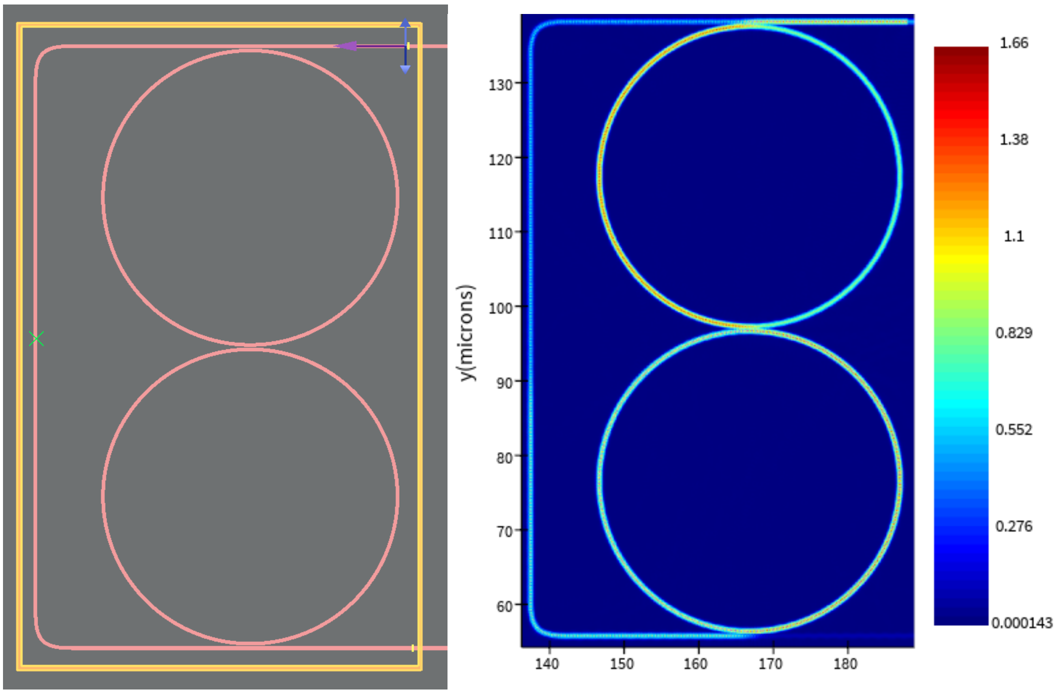
Figure 6: Power spectra at different monitors in the ring reflector with gap of (a) 100nm and (b) 200nm. 'East' and 'west' refer to monitors at the leftmost and rightmost parts of the ring, as seen in Fig. 4.

target wavelength (μm)	radius (μm)
1.50	19.09
1.55	20.15
1.60	21.32

Table 2: Table of target resonant wavelengths with $m = 200$ and calculated ring resonator radii.

3.4 Optical reflector

The key component in the optical refelctor is the coupled ring resonator: see Fig. 7. Light on resonance travels an S-shaped path through the device, exiting back the way it came. varFDTD simulations how this behaviour. Much like the dual-bus ring resonators, two waveguide-ring gaps are chosen: 100nm and 200nm. These affect the efficiency of the reflection in the expected way: see Fig. 8. The ring radii are chosen as 20.15 mum, with target wavelength 1.55 mum.



(a) (b)

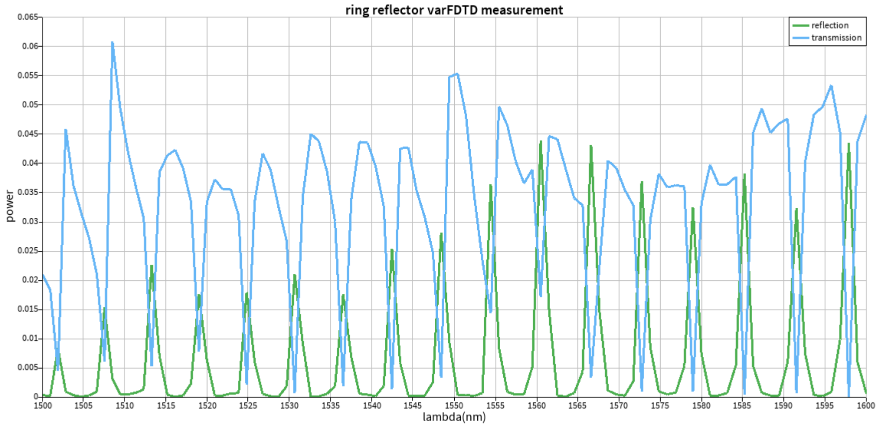
Figure 7: (a) structure of coupled ring resonator in Lumerical MODE. (b) varFDTD simulated electric field in the device close to ring resonance. Note significantly reduced field at output (bottom right).

4 Fabrication

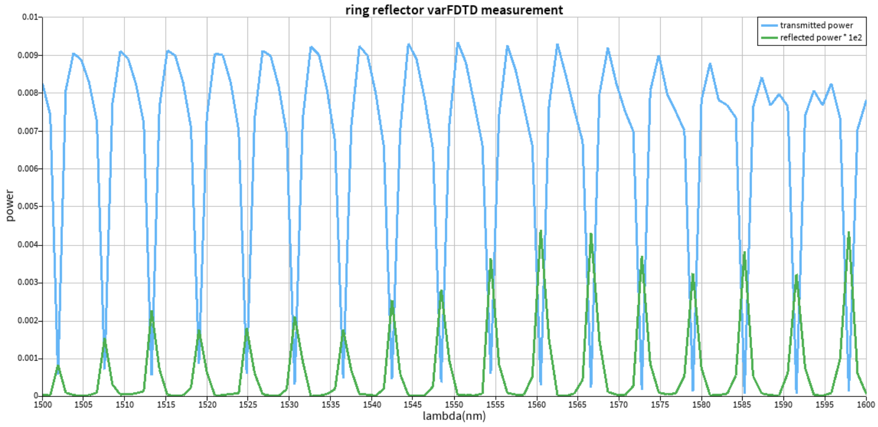
Fabrication was performed at Applied Nanotools.

4.1 NanoSOI process

The photonic devices were fabricated using the NanoSOI MPW fabrication process by Applied Nanotools Inc. (http://www.appliednt.com/nanosoi; Edmonton, Canada) which is based on direct-write 100 keV electron beam lithography technology. Silicon-on-insulator wafers of 200 mm diameter, 220 nm device thickness and 2 µm buffer oxide thickness are used as the base material for the fabrication. The wafer was pre-diced into square substrates with dimensions of 25x25 mm, and lines were scribed into the



(a)



(b)

Figure 8: Transmission and reflection spectra for coupled rings with gap (a) 100nm and (b) 200nm. Note in (b): the reflection signal has been artifically amplified by a factor of 100 to be visible on the plot.

substrate backsides to facilitate easy separation into smaller chips once fabrication was complete. After an initial wafer clean using piranha solution (3:1 H2SO4:H2O2) for 15 minutes and water/IPA rinse, hydrogen silsesquioxane (HSQ) resist was spin-coated onto the substrate and heated to evaporate the solvent. The photonic devices were patterned using a JEOL JBX-8100FS electron beam instrument at The University of British Columbia. The exposure dosage of the design was corrected for proximity effects that result from the backscatter of electrons from exposure of nearby features. Shape writing order was optimized for efficient patterning and minimal beam drift. After the e-beam exposure and subsequent development with a tetramethylammonium sulfate (TMAH) solution, the devices were inspected optically for residues and/or defects. The chips were then mounted on a 4" handle wafer and underwent an anisotropic ICP-RIE etch process using chlorine after qualification of the etch rate. The resist was removed from the surface of the devices using a 10:1 buffer oxide wet etch, and the devices were inspected using a scanning electron microscope (SEM) to verify patterning and etch quality. A 2.2 µm oxide cladding was deposited using a plasma-enhanced chemical vapour deposition (PECVD) process based on tetraethyl orthosilicate (TEOS) at 300ºC. Reflectrometry measurements were performed throughout the process to verify the device layer, buffer oxide and cladding thicknesses before delivery.

5 Measurement description

To characterize the devices, a custom-built automated test setup [2, 6] with automated control software written in Python was used [3]. An Agilent 81600B tunable laser was used as the input source and Agilent 81635A optical power sensors as the output detectors. The wavelength was swept from 1500 to 1600 nm in 10 pm steps. A polarization maintaining (PM) fibre was used to maintain the polarization state of the light, to couple the TE polarization into the grating couplers [4]. A 90º rotation was used to inject light into the TM grating couplers [4]. A polarization maintaining fibre array was used to couple light in/out of the chip [5].

6 Measurement analysis

Analysis was performed using Lumerical MODE to perform a corner analysis of the group index of the waveguide - 500nm width, 220 nm thickness, TE polarisation - followed by analysis of measurement data using Python.

6.1 Corner analysis

Corner analysis was performed assuming a range of widths from 470nm to 510nm, and a range of thicknesses from 215.3nm to 223.1nm. Eigenmode simulations were used to extract compact model parameters for the effective refractive index:

$$n_{eff} = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2$$

Group index at λ_0 was calculated via:

$$\begin{aligned} n_g &= \left(n_{eff} - \lambda \frac{dn_{eff}}{d\lambda}\right)\Big|_{\lambda=\lambda_0} \\ &= n_1 - n_2\lambda_0 \end{aligned}$$

The maximum and minimum n_g calculated were 4.279 and 4.198.

6.2 Python analysis

Grating coupler response was extracted from measurement of a simple loopback. The output gain was fit with an 8th order polynomial, and this fit was then subtracted from the device gain responses, plus a factor of 5.5 dB to fix the maximum gain level to 0dB. From this, the FSR was extracted using the `find_peaks` function in SciPy's `signal` package. These FSRs, and calculated group indices, are presented in Table. 3.

Device	FSR (nm)	n_g
MZI1	4.29	2.3
MZI2	8.20	2.8
Ring reflector 1	11.4	1.7
Ring reflector 2	9.40	2.0
Demultiplexer 1	7.25	2.5
Demultiplexer 2	13.3	1.3

Table 3: Table of FSR and group indices for devices on the chip. Demultiplexer values are calculated from the first output associated with the ring resonator with length 21.32 μm.

It is clear that the calculated group indices are well outside the range calculated via the corner analysis. The demultiplexers use the same ring resonator path length, as do the ring reflectors, with the difference between the devices only being the coupling gaps between waveguides and ring resonators. As such, it is unclear why the differences in FSR should be so significant, especially in the case of the demultiplexers.

7 Acknowledgments

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8 References

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[3] http://siepic.ubc.ca/probestation, using Python code developed by Michael Caverley.

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